

Student Academic Performance Predictor

Machine Learning Project Report | Regression Analysis | 4,000 Student Records

PRESENTED BY

B. Mohammed Afraz Hussain Basha • P.S.H. Kishan Kesava • D. Rama Sri Charan

Internship Project — Eduexpose

4,000 Student Records	91.6% R ² Score	3.99 MAE	5.03 RMSE	4 Models Compared
--------------------------	-------------------------------	-------------	--------------	----------------------

1. Introduction

This report presents the development of a supervised machine learning regression model to predict student exam scores based on a combination of behavioural and demographic features. The dataset consists of 4,000 student records collected across urban and rural regions, capturing variables such as weekly study hours, attendance percentage, tutoring status, gender, region, and parental education level.

The primary objective is to identify the most influential predictors of academic performance and to build a model capable of accurate score prediction that can support educators in early identification of at-risk students.

1.1 Project Scope

Project Type	Supervised ML — Regression
Target Variable	Exam_Score (continuous, range 17–100)
Dataset Size	4,000 records, 7 columns (6 features + 1 target)
Tools Used	Python, scikit-learn, pandas, matplotlib, seaborn, SHAP, Jupyter Notebook
Environment	Google Colab
Best Model	Linear Regression — Test R ² = 0.9156, MAE = 3.99, RMSE = 5.03

2. Dataset Description

The dataset **SAP-4000.csv** contains 4,000 student records after duplicate removal. Each record contains 6 input features and 1 target variable. Data was sourced synthetically to represent a diverse student population across regions.

2.1 Feature Summary

Feature	Type	Range / Values	Notes
HoursStudied/Week	Numerical	0 – 16 hrs	Strongest predictor (r = 0.82)
Attendance(%)	Numerical	50 – 100%	Weak correlation (r = 0.24)
Gender	Categorical	Male / Female	~51% Female, 49% Male
Tutoring	Categorical	Yes / No	70% students: No tutoring
Region	Categorical	Urban / Rural	60% Urban, 40% Rural

Parent Education	Categorical	Primary / Secondary / Tertiary	422 missing values handled
Exam_Score (TARGET)	Numerical	17 – 100	Peaks in 60–80 range

2.2 Data Quality

- No duplicate records after cleaning — final shape: (4000, 7)
- Parent Education had 422 missing values (10.5%) — imputed using the most frequent strategy within the preprocessing pipeline
- All other columns had zero missing values
- No outlier removal was performed; the score range (17–100) is expected and valid

2.3 Correlation Analysis

A Pearson correlation matrix was computed for the three numerical features. The key finding is a strong positive correlation between **HoursStudied/Week** and **Exam_Score** ($r = 0.819$), indicating a near-linear relationship. **Attendance(%)** shows a much weaker correlation ($r = 0.239$), confirming that attendance alone is insufficient to predict performance.

	HoursStudied/Week	Attendance(%)	Exam_Score
HoursStudied/Week	1.000000	0.003607	0.819812
Attendance(%)	0.003607	1.000000	0.238570
Exam_Score	0.819812	0.238570	1.000000

3. Methodology

3.1 Preprocessing Pipeline

A scikit-learn Pipeline was used to wrap all preprocessing and model steps, ensuring that transformations are fit on training data only and applied to test data — preventing data leakage.

- **Numerical features** (HoursStudied/Week, Attendance%): StandardScaler — zero mean, unit variance
- **Categorical features** (Gender, Tutoring, Region, Parent Education): OneHotEncoder — drop='first' to avoid multicollinearity
- **Missing values** in Parent Education: SimpleImputer with strategy='most_frequent' inside the pipeline
- Train/test split: 80% / 20% with random_state=42

3.2 Models Trained

Four regression models were evaluated within the preprocessing pipeline:

- **Linear Regression** — baseline, assumes linear relationships
- **Decision Tree Regressor** — non-linear, hierarchical splits
- **Random Forest Regressor** — ensemble of decision trees, reduces overfitting
- **Gradient Boosting Regressor** — sequential ensemble, optimises residuals

3.3 Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was applied to the Random Forest model to find optimal hyperparameters:

Search Parameters	$n_estimators \in \{100, 200\}$, $max_depth \in \{5, 10, None\}$
Best Parameters	$max_depth = 10$, $n_estimators = 200$
Best CV Score	$R^2 = 0.8947$ (5-fold average)
Scoring Metric	R^2 (coefficient of determination)

4. Results & Model Evaluation

4.1 Model Comparison

Model	Train R^2	Test R^2	MAE	RMSE	CV R^2 (5-fold)
★ Linear Regression	0.9061	0.9156	3.99	5.03	0.9073
Gradient Boosting	0.9202	0.9114	—	—	0.9056
Random Forest (tuned)	0.9846	0.8984	—	—	0.8915
Decision Tree	0.9997	0.8069	5.88	7.60	0.8012

★ Best model | Red values indicate weak generalisation (Decision Tree overfitting)

4.2 Key Findings

- **Linear Regression achieved the best test performance** with $R^2 = 0.9156$, MAE = 3.99, RMSE = 5.03 — outperforming all more complex models.
- **Decision Tree severely overfits**: Train $R^2 = 0.9997$ vs Test $R^2 = 0.8069$ — a gap of ~0.19 points.
- **Gradient Boosting is the second-best** generaliser (Test $R^2 = 0.9114$) and shows the most stable CV score.
- **Random Forest performs reasonably** after tuning (Test $R^2 = 0.8984$) but cannot surpass the simpler linear model.
- The dominance of Linear Regression confirms that the feature-target relationship is **fundamentally linear**, driven primarily by HoursStudied/Week.

4.3 Learning Curve Analysis

The learning curve for the best model shows that training score starts at 0.922 with ~320 examples and converges toward the validation score (~0.908) as training size reaches 3,200. The validation score monotonically increases from 0.905 to 0.908. The narrow and converging gap between training and validation scores confirms **no significant overfitting** — the model generalises well to unseen data and would benefit modestly from additional training examples.

4.4 Residual Diagnostics

The residual plot and prediction error distribution confirm the following:

- Residuals are scattered randomly around zero — **no systematic bias**
- Prediction error distribution is **approximately normal**, centred at zero
- Majority of predictions fall within **±5 marks** of the true score
- Outlier tails extend to ±15–17 marks for edge cases
- Slight heteroscedasticity visible at high predicted scores (100+) — expected due to score ceiling effect

5. Feature Importance — SHAP Analysis

SHAP (SHapley Additive exPlanations) values were computed using the **LinearExplainer** on 100 subsampled test observations. SHAP quantifies each feature's marginal contribution to individual predictions.

5.1 SHAP Results Summary

Feature	Impact Level	Interpretation
HoursStudied/Week	DOMINANT	SHAP values span -35 to +25. Each additional study hour strongly shifts predicted
Attendance(%)	Moderate	Secondary positive driver. Effect is much smaller than study hours.
Tutoring (Yes/No)	Small	Small positive signal. Tutored students score marginally higher.
Parent Education	Minimal	Near-zero SHAP values across all levels (Primary/Secondary/Tertiary).
Gender	Negligible	Essentially zero contribution. Male/Female SHAP values overlap.
Region	Negligible	Urban vs Rural has no meaningful predictive effect.

5.2 Key Insight

The SHAP beeswarm plot (see `graphs/feature_importance.png`) reveals that **HoursStudied/Week accounts for the vast majority of predictive variance**. Students who study more than 10 hours/week consistently receive SHAP values above +15, while students studying less than 2 hours/week receive SHAP values below -20. This dominance confirms that study behaviour is the primary lever for improving academic outcomes — not demographic characteristics.

6. Conclusions

6.1 Summary

- Weekly study hours is the **single most powerful predictor** of exam score, with a Pearson correlation of $r = 0.82$ and dominant SHAP values.
- Linear Regression achieves the best overall performance** (Test $R^2 = 0.9156$, MAE = 3.99, RMSE = 5.03) — surpassing all more complex models including Random Forest and Gradient Boosting.
- The model is **well-generalised with no significant overfitting**, as evidenced by the converging learning curve and symmetric error distribution.
- Demographic features including gender, region, and parent education have **negligible predictive power** in this dataset.
- The preprocessing pipeline with StandardScaler and OneHotEncoder correctly handles both numerical and categorical features without data leakage.

6.2 Future Work

- Collect richer features: sleep hours, motivation scores, socioeconomic index, teacher quality ratings
- Deploy the model as a web application using Flask or Streamlit for real-time predictions by educators
- Explore deep learning approaches (MLP, Transformer) on a larger dataset (10,000+ records)

- Build an early-warning alert system that flags at-risk students based on predicted scores
- Investigate non-linear feature engineering (interaction terms, polynomial features) to further improve accuracy

7. Appendix

7.1 Cross-Validation Results (5-fold R²)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Linear Regression	0.9174	0.9032	0.9003	0.9114	0.9040	0.9073
Gradient Boosting	0.9173	0.9051	0.8974	0.9091	0.8990	0.9056
Random Forest	0.9038	0.8935	0.8828	0.8919	0.8853	0.8915
Decision Tree	0.8138	0.8129	0.7921	0.7853	0.8017	0.8012

7.2 Project File Structure

SAP-4000.csv	Raw dataset — 4,000 student records, 7 columns
Student_Performance.ipynb	Main Jupyter Notebook (EDA + full ML pipeline)
student_performance_model.pkl	Saved best model (Linear Regression pipeline)
Model_Comparison.csv	R², MAE, RMSE results for all 4 models
Predictions.csv	Actual vs Predicted values on test set
requirements.txt	Python package dependencies
README.md	Project documentation and usage guide
report.pdf	This project report
presentation.pptx	Internship presentation slides (8 slides)

7.3 How to Run the Model

Install dependencies:

```
pip install -r requirements.txt
```

Load the model and make a prediction:

```
import joblib, pandas as pd
model = joblib.load("student_performance_model.pkl")
sample = pd.DataFrame([{"Gender": "Female", "HoursStudied/Week": 10.0,
                        "Tutoring": "Yes", "Region": "Urban",
                        "Attendance(%)": 85, "Parent Education": "Secondary"}])
print(f"Predicted Score: {model.predict(sample)[0]:.2f}")
```

End of Report — Student Academic Performance Predictor